

# Market Basket Analysis

## Insights & Recommendations Report

### 1. Project Objective

The objective of this project is to analyse retail transaction data using Market Basket Analysis (Apriori Algorithm) to identify product associations and generate actionable business insights that can improve cross-selling, bundling strategies, and overall revenue performance.

### 2. Key Findings Summary

Based on the Apriori algorithm and association rule mining, the following major patterns were identified:

- i. Customers frequently purchase complementary products together, especially within the same category (e.g., party supplies and decorative items).
- ii. Strong association rules were observed between matching or themed products (e.g., wooden star Christmas & wooden heart Christmas).
- iii. Several product pairs showed high lift values indicating strong cross-selling potential.
- iv. Seasonal and gifting items showed significantly higher co-purchase frequency compared to regular items.
- v. Customers tend to purchase multiple design or colour variants of similar products in a single transaction.

### **3. Business Insights**

The analysis reveals important strategic insights for the business:

- There is strong cross-selling potential among decorative and party-related items.
- Customers respond positively to theme-based or matching products.
- High-lift product combinations indicate customer purchase intent rather than random coincidence.
- Certain low-frequency items become valuable when bundled with popular products.
- Basket size can be increased through strategic placement and promotional offers.

### **4. Actionable Business Recommendations**

Based on the insights, the following strategic actions are recommended:

#### **a. Product Bundling Strategy:**

Create combo packs for highly associated products.

Example: Party decoration bundles, Christmas themed packs, gifting combos.

#### **b. Store Layout Optimization:**

Place frequently co-purchased products near each other to encourage impulse buying and increase basket value.

#### **c. Cross-Selling Promotions**

Offer targeted discounts such as:

“Buy Product A and get 5% off on Product B.”

#### **d. Online Recommendation Engine**

For e-commerce platforms, implement “Customers who bought this also bought...” recommendations using association rules.

#### **e. Inventory Planning**

Ensure sufficient stock of highly associated products during peak seasons to avoid lost sales opportunities.

### **5. Expected Business Impact**

If implemented effectively, the recommendations can:

- Increase Average Order Value (AOV)
- Improve cross-selling conversion rate
- Enhance customer shopping experience
- Optimize inventory management
- Drive higher overall revenue

Market Basket Analysis enables data-driven decision-making instead of assumption-based product placement.

### **6. Limitations**

- Analysis is based only on historical transaction data.
- Pricing, discounts, and profit margins were not considered.
- Customer segmentation was not included.
- Seasonal variation may impact association strength.

### **7. Future Scope**

To enhance this analysis further:

- Implement FP-Growth algorithm for faster processing.
- Combine MBA with customer segmentation.
- Perform time-based association analysis.
- Develop a real-time recommendation system.

